

No cumulative finite-Blaschke basis of the disk algebra

A literature-implied full answer to Question 7.4 of arXiv:2507.13121

Status. Literature-implied full negative answer. The decisive obstruction is Theorem 4 of Ivanov–Shekhtman (2001), which predates the source question. The reduction below includes repeated zeros, which ordinary point interpolation does not cover verbatim.

Source and exact question. The source is E. Fricain, J. Mashreghi, M. Nasri, and M. Ostermann, *Schauder Basis with Finite Blaschke Products*, arXiv:2507.13121v2. It sets

$$B_0 = 1, \quad B_n(z) = \prod_{k=1}^n b_{\lambda_k}(z), \quad b_{\lambda}(z) = \frac{\lambda - z}{1 - \bar{\lambda}z},$$

and calls (λ_n) non-Blaschke when $\sum_n (1 - |\lambda_n|) = \infty$.

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Nevertheless, this does not solve the question for all non-Blaschke sequences $(\lambda_n)_{n \geq 1}$ satisfying $\sup_n |\lambda_n| = 1$. In particular, this leaves unknown the answer to the following intriguing question.

Question 7.4. Does there exist a Schauder basis of $\mathcal{A}(\mathbb{D})$ of the form $(B_n)_{n \geq 1}$ where $(\lambda_n)_{n \geq 1}$ is a non-Blaschke sequence in $\mathbb{D}$?

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Figure 1: Question 7.4 on printed page 16 of arXiv:2507.13121v2.

The displayed indexing starts at 1. Literally, that version has the immediate negative answer: every B_n , $n \geq 1$, vanishes at λ_1 , so its closed span is contained in the proper ideal $\{f \in \mathcal{A}(\mathbb{D}) : f(\lambda_1) = 0\}$. We prove the stronger intended statement for the complete family $(B_n)_{n \geq 0}$.

Idea of proof. The tail of the proposed basis is not an arbitrary subspace: after the first N terms it is exactly the principal ideal $B_N \mathcal{A}(\mathbb{D})$. Thus the N th basis projection is the Hermite interpolation projection at the zeros of B_N , counted with multiplicity. Split repeated nodes into nearby distinct nodes. The associated Lagrange projections converge in operator norm to the Hermite projection. A putative basis would therefore yield point-interpolation projections converging to the identity, which Ivanov–Shekhtman proved impossible in the disk algebra.

Theorem. *For every non-Blaschke sequence $(\lambda_n) \subset \mathbb{D}$, repetitions allowed, the cumulative family $(B_n)_{n \geq 0}$ is not a Schauder basis of $\mathcal{A}(\mathbb{D})$. Consequently Question 7.4 has a negative answer under either its literal or its intended indexing.*

Proof. Assume that $(B_n)_{n \geq 0}$ is a Schauder basis of $\mathcal{A}(\mathbb{D})$, and let

$$S_N \left(\sum_{n \geq 0} a_n B_n \right) = \sum_{n=0}^{N-1} a_n B_n, \quad E_N = \text{span}\{B_0, \dots, B_{N-1}\}.$$

Then $S_N f \rightarrow f$ for every $f \in \mathcal{A}(\mathbb{D})$ and

$$\ker S_N = \overline{\text{span}}\{B_n : n \geq N\}.$$

Write $B_{N+k} = B_N C_k$, where $C_0 = 1$ and C_k is the cumulative product formed from $\lambda_{N+1}, \dots, \lambda_{N+k}$. Removing finitely many terms preserves the non-Blaschke condition. The source paper’s completeness criterion therefore gives $\overline{\text{span}}\{C_k : k \geq 0\} = \mathcal{A}(\mathbb{D})$. Multiplication by B_N is an isometry of $\mathcal{A}(\mathbb{D})$ onto the closed ideal $B_N \mathcal{A}(\mathbb{D})$, since $|B_N| = 1$ on $\partial \mathbb{D}$. Hence

$$\ker S_N = B_N \mathcal{A}(\mathbb{D}). \tag{1}$$

Let $\alpha_1, \dots, \alpha_s$ be the distinct zeros of B_N , with multiplicities $m_1, \dots, m_s$; thus $\sum_j m_j = N$. Divisibility by B_N is equivalent to vanishing of the jets

$$f^{(r)}(\alpha_j), \quad 1 \leq j \leq s, \quad 0 \leq r < m_j.$$

Equation (1) says precisely that $S_N f$ is the unique element of E_N sharing all these jets with f . (Uniqueness also follows from $E_N \cap B_N \mathcal{A}(\mathbb{D}) = \{0\}$.) Thus S_N is a confluent, or Hermite, interpolation projection.

For this fixed N , replace the m_j copies of each α_j by m_j distinct points $\alpha_{j,r}(t) \in \mathbb{D}$ tending to α_j as $t \downarrow 0$. For all sufficiently small t , interpolation at the resulting N distinct nodes is unisolvent on E_N ; denote its projection by $L_{N,t} : \mathcal{A}(\mathbb{D}) \rightarrow E_N$. The confluence lemma below gives

$$\|L_{N,t} - S_N\|_{\mathcal{A}(\mathbb{D}) \rightarrow \mathcal{A}(\mathbb{D})} \longrightarrow 0. \quad (2)$$

Choose t_N so small that the norm in (2) is less than $1/N$, and put $L_N = L_{N,t_N}$. Each L_N is a projection determined only by the values of its input on a finite set $Z_N \subset \mathbb{D}$. Moreover, for every $f \in \mathcal{A}(\mathbb{D})$,

$$\|L_N f - f\|_\infty \leq \|(L_N - S_N)f\|_\infty + \|S_N f - f\|_\infty \leq \frac{\|f\|_\infty}{N} + \|S_N f - f\|_\infty \longrightarrow 0.$$

This contradicts Ivanov–Shekhtman, Theorem 4: for arbitrary finite sets $Z_N \subset \mathbb{D}$ and arbitrary projections L_N depending only on values on Z_N , there is an $f \in \mathcal{A}(\mathbb{D})$ for which $L_N f$ does not converge to f . The assumed Schauder basis cannot exist. $\square$

Confluence lemma. *With the notation above, $L_{N,t} \rightarrow S_N$ in operator norm as the distinct nodes coalesce to the zeros of B_N with their prescribed multiplicities.*

Proof. Let $V_t : \mathcal{A}(\mathbb{D}) \rightarrow \mathbb{C}^N$ be the value map at the split nodes. Within each cluster apply the standard Newton divided-difference change of coordinates D_t to V_t . Because all limiting nodes lie strictly inside $\mathbb{D}$, Cauchy estimates on fixed small disks show, uniformly for $\|f\|_\infty \leq 1$, that $D_t V_t f$ converges to the jet vector

$$Jf = (f^{(r)}(\alpha_j)/r!)_{j,r}.$$

The same convergence holds for the restrictions to the fixed finite-dimensional space E_N . The restriction $J|_{E_N}$ is invertible, since its kernel is $E_N \cap B_N \mathcal{A}(\mathbb{D}) = \{0\}$. Consequently

$$L_{N,t} = ((D_t V_t)|_{E_N})^{-1} D_t V_t \longrightarrow (J|_{E_N})^{-1} J = S_N$$

in operator norm, by continuity of inversion in finite dimensions. $\square$

Literature and novelty scope. For pairwise distinct λ_n , the contradiction is essentially immediate from the 2001 theorem: $S_N f$ interpolates f at $\lambda_1, \dots, \lambda_N$. The confluence argument closes the repetition case allowed by the source definition. The run therefore records this as a literature-implied answer, not as a new independent solution. A bounded search through 2026-08-11 found no later paper needed for the conclusion.

Review recommendation. The mathematical reduction is short and checkable, but the status should remain *literature-implied*: verify the cited formulation of Ivanov–Shekhtman Theorem 4, then notify the source authors of the pre-existing interpolating-basis obstruction and the confluent-node extension.

References

- [1] E. Fricain, J. Mashreghi, M. Nasri, and M. Ostermann, *Schauder Basis with Finite Blaschke Products*, arXiv:2507.13121v2 (2026).
- [2] I. V. Ivanov and B. Shekhtman, *Linear discrete operators on the disk algebra*, Proc. Amer. Math. Soc. **129** (2001), 1987–1993, doi:10.1090/S0002-9939-00-05774-9.